

INVENTION DISCLOSURE FORM

Applicant Name-Marwadi University

1. Particulars of Inventors

Mr./Ms/Dr.	Name (Full)	Department(with division)	Designation	Mobile No.	Email
Mr.	Dhyey Ajaybhai Raja	CSE-AI&ML PBL(3EG1)	Student	9408301872	dhyey.raja153148@marwadiuniversity.a

Hybrid CNN–Transformer Architecture for Adversarial Image Classification

1. In 100 words or less, please provide an abstract or summary of the invention:

Abstract

This invention presents a new image classification system that combines CNN and Transformer technology. CNN helps the system identify important details such as shapes, edges, and textures, while the Transformer understands the relationship between different parts of an image. The system is designed to recognize images accurately even when small changes are intentionally made to confuse the AI. By combining these two technologies, the proposed system aims to improve the accuracy, reliability, and resistance of image classification against adversarial attacks. It can be useful in areas such as security, healthcare, surveillance, and autonomous vehicles.

1. Detail description of the invention:(Answer to all below are required in detail)

1. Problem the invention is solving

Nowadays, image classification is used in many areas such as security, healthcare, vehicles, and surveillance. Normally, an AI model can look at an image and identify whether it is a cat, dog, car, person, or another object.

However, there is a problem called an adversarial attack. In this type of attack, someone makes very small changes to an image. These changes may not be easily noticed by a person, but they can confuse an AI model and make it give the wrong answer.

For example, an image of a dog may still look like a dog to us, but after making some small changes, an AI model may incorrectly identify it as a cat.

The main purpose of our invention is to reduce this problem. We propose a model that combines a CNN and a Transformer. The CNN is useful for finding small details in an image, while the Transformer helps the model understand the overall image and the relationship between different parts. By using both, the system is intended to make image classification more reliable and more resistant to adversarial changes.

- 1. **General Utility/application of the invention**

The proposed system can be useful wherever computers need to identify images correctly.

Some possible applications are:

1. Security and Surveillance

The system can be used in CCTV and security systems to recognize people, objects, or suspicious activities. Its resistance to modified images can make such systems more reliable.

2. Medical Imaging

The system can be used for classifying medical images such as X-rays, CT scans, or other medical images. An image classification system can help doctors by providing additional information.

3. Autonomous Vehicles

Self-driving or driver-assistance systems need to recognize objects such as cars, pedestrians, traffic signs, etc. A robust image classification system can help reduce errors caused by manipulated or unusual images.

4. Face Recognition

The system can be used in biometric and face-recognition applications where accurate identification is important.

5. Industrial Inspection

Factories can use image classification to identify damaged or defective products automatically.

6. Agriculture

The system can also be used to identify plant diseases or classify different types of crops from photographs.

- 1. **Advantages of the invention disclosing about the increased efficiency/efficacy**

The main advantage of this invention is that it combines two different types of deep-learning models instead of depending only on one model.

The CNN is good at identifying local details such as edges, textures, shapes, and patterns. The Transformer is good at relationships between different parts of an image and understanding more overall information.

at relationships between different parts of an image and understand more overall information.

Some important advantages are:

- It combines the useful features of CNN and Transformer models.
- It can improve image classification performance.
- It is designed to handle adversarially modified images better.
- It can reduce the possibility of incorrect classification caused by small image changes.
- It can be used with different image datasets.
- The model can be modified according to the requirements of a particular application.
- It can be useful in applications where reliable image recognition is important.

1. Best way of using the invention as well as possible variants

The proposed system can be developed using Python and a deep-learning framework such as PyTorch or TensorFlow.

First, an image dataset is selected. The images are then resized and prepared so that they can be given as input to the model.

After preprocessing, the image is passed through the CNN part of the system. The CNN extracts useful information such as edges, shapes, textures, and other visual features.

The extracted information is then provided to the Transformer part. The Transformer helps the system understand the relationship between different areas of the image.

After combining the information obtained from both parts, the final classification layer predicts the class of the image.

To study adversarial robustness, modified versions of the images can also be created using adversarial attack methods. The model can then be trained or tested using these images to check how well it performs under adversarial conditions.

Possible Variants

The basic idea can be changed or improved in different ways.

For example:

- Different CNN models such as ResNet or EfficientNet can be used.
- Different Transformer models can be used depending on the requirements.
- Different image datasets can be used.
- Different adversarial attack methods can be tested.
- Different defence or adversarial-training techniques can be added.
- The model can be designed for small devices or cloud-based systems.
- The system can be adapted for medical, security, industrial, or agricultural images.

1. Working of invention along with Drawing, schematics and flow diagrams if required with clear explanations

The working of the proposed system can be understood in a simple step-by-step manner.

Step 1: Input Image

First, an image is given to the system.

For example:

Input → Image of a Dog

Step 2: Image Preprocessing

The image is prepared before sending it to the model. This may include resizing the image and normalizing values.

This makes the image suitable for processing by the deep-learning model.

Step 3: CNN Feature Extraction

The processed image is passed through the CNN.

The CNN looks for important visual details such as:

- Edges
- Shapes
- Textures
- Patterns
- Small objects or parts of objects

For example, it may detect features such as the dog's ears, eyes, fur, and body shape.

Step 4: Transformer Processing

The features obtained from the CNN are then given to the Transformer.

The Transformer looks at the relationship between different parts of the image. This helps the model understand the image more completely rather than looking at individual details separately.

Step 5: Combining the Features

The information obtained from the CNN and Transformer is combined.

In simple words:

CNN tells the system about important details.

Transformer helps the system understand how those details are related.

Together, they provide information that can be used for classification.

Step 6: Classification

The combined information is given to the final classification layer.

The system then produces a prediction such as:

Dog → 95%

Cat → 3%

Horse → 2%

The class with the highest probability becomes the final prediction.

Step 7: Testing Against Adversarial Images

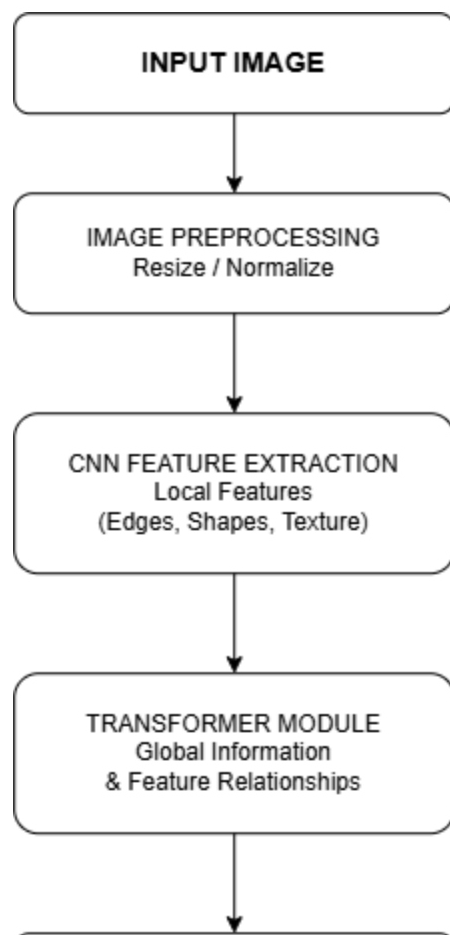
The system can also be tested with images that have been intentionally modified to confuse the AI.

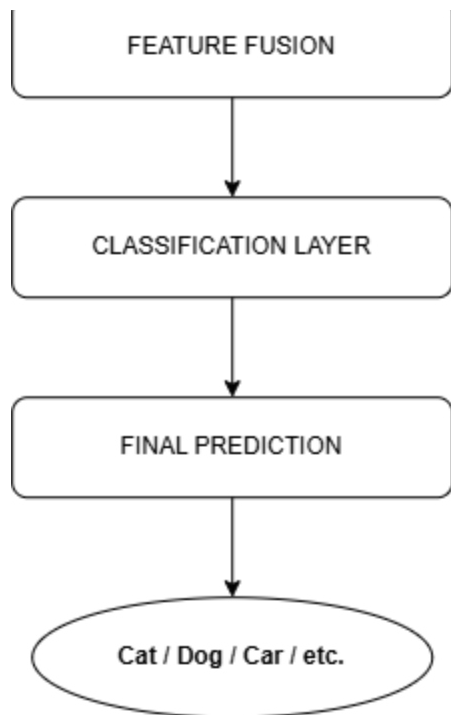
For example:

Original image → Dog → Correct prediction

Modified/adversarial image → Dog → Model should try to correctly identify it as Dog

The performance on normal and adversarial images can then be compared.





1. **Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search results**

No.

A primary patent search has not yet been conducted for the proposed invention titled “Hybrid CNN–Trans Architecture for Adversarial Image Classification.”

1. **Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/prior art in the market)**

Today, image classification is commonly used in many applications such as face recognition, medical image systems, CCTV, and self-driving vehicles. Different types of AI models are already available for identifying images.

CNN is one of the most commonly used methods for image classification. It is good at finding small detail image, such as edges, shapes, textures, and patterns. However, CNN models can sometimes be fooled when an image is slightly changed by an attacker.

Another approach is the Transformer model. It can look at different parts of an image and understand how they are related to each other. Transformer-based models have also become popular for image classification.

Researchers have also developed different methods to protect AI models from adversarial attacks. These methods aim to teach the model to handle images that have been intentionally changed.

Even though CNNs, Transformers, and adversarial-defence methods already exist, there is still a need for combining these technologies. Our proposed work focuses on combining CNN and Transformer so that it can use both detailed image information and overall image information while trying to remain reliable when the image is modified.

1. List out the known ways about how others have tried to solve the same or similar problems? disadvantages of these approaches. In addition, please identify any prior art documentation or material that explains or provides examples of such prior art efforts.

Different methods have already been used to improve image classification and protect AI models from adversarial attacks. Some common approaches and their limitations are given below.

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	Traditional CNN-based image classification is commonly used to recognize and classify images.	CNNs are good at finding small image features, but they may not capture relationships between distant parts of an image very well. They can also be affected by adversarial changes.	Our approach combines CNN and Transformer so that local features and wider image information can be used together. Transformer can be used to capture global dependencies.
2	Adversarial training is used to train a model using both normal and adversarial images.	It can improve resistance to attacks, but training can take more time and computing resources. In some cases, improving robustness can also affect normal-image accuracy.	Our proposed system combines adversarial training with the hybrid CNN-Transformer model to improve robustness while maintaining useful image feature classification.

1. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

Technical Features and Elements of the Invention

The proposed invention is a Hybrid CNN–Transformer Architecture for Adversarial Image Classification. The main technical features and elements of the system are described below in the order in which they work.

S. No.	Technical Feature / Element	Description
1	Input Image	The process begins with an input image, which is fed into the Hybrid CNN–Transformer Architecture for Adversarial Image Classification.

The system first receives an image that needs to be classified. The image may be a normal image or been modified by an adversarial attack.

2	The input image is prepared before being given to the model. This may include resizing normalizing its pixel values so that the model can process it properly.	
Image Preprocessing		
3	CNN Feature Extraction	Extraction
The image is passed through the CNN part of the model. The CNN identifies useful visual details such shapes, textures, and patterns.		
4	Feature Representation	The features obtained from the CNN are converted into a suitable representation that c processed by the Transformer.
5	The Transformer processes the extracted features and looks at the relationships between parts of the image. This helps the model understand the image at a broader level.	
Transformer Module		
6	Attention Mechanism	The Transformer uses attention to give importance to different parts of the input feature the model focus on the information that is more useful for classification.
7	Feature Combination	Information obtained from the CNN and Transformer is combined so that the final model can use both details and broader image information.
8	Adversarial Training / Defense	The system can be trained or tested using adversarially modified images. This helps ev improve the model's ability to handle images that are intentionally changed to confuse t
9	Classification Layer	

		The combined features are passed to the final classification layer. The model calculates the likelihood belonging to different classes.
10	Final Prediction	The class with the highest predicted probability is selected as the final result, such as C or another class depending on the dataset.
11	Performance Evaluation	The model is evaluated using measures such as accuracy, precision, recall, F1-score, and robustness on adversarial images.
12	Robustness Comparison	The performance of the proposed hybrid model can be compared with a normal CNN or Transformer model to understand whether the proposed approach provides better resistance to adversarial changes.

Description of the Invention from Start to End

The system starts by taking an image as input. The image is first resized and normalized so that it has the correct format for the model. The prepared image is then passed through the CNN, which extracts important details such as edges, shapes, textures, and patterns.

The extracted information is then provided to the Transformer. The Transformer examines the relationships between different parts of the image and helps the model understand the overall information contained in the image.

The information from the CNN and Transformer is then combined. This combined information is passed to the final classification layer, which produces the final prediction.

To check the robustness of the system, the model can also be tested with adversarial images. These are images that have been slightly modified with the intention of confusing the AI model. The results from normal and adversarial images are compared to see how well the proposed system performs.

Therefore, the complete process can be summarized as:

Input Image → Preprocessing → CNN Feature Extraction → Transformer Processing → Feature Combination → Adversarial Training/Testing → Classification → Final Prediction → Performance Evaluation

The main technical idea of the invention is to use the CNN and Transformer together instead of relying on a single type of model. The CNN provides detailed visual information, while the Transformer provides information about the relationships between different parts of the image. The combination is intended to provide a more reliable and robust adversarial image classification.

1. **List out the features of your invention which are believed to be new and distinguish them over technology.**

Features believed to be new and distinguishing over the closest technology

1. Hybrid CNN–Transformer Architecture

The invention combines CNN and Transformer technologies in a single image-classification architecture.

1. Local and Global Feature Understanding

The CNN extracts local visual features such as edges, shapes, textures, and patterns, while the Transformer captures global relationships between different parts of the image.

1. Integrated Feature Combination

The features obtained from the CNN and Transformer are combined before classification, allowing the system to utilize both detailed and broader image information.

1. Adversarial Image Classification

The proposed system is designed to classify images that may have been intentionally modified to confuse traditional models.

1. Adversarial Robustness Evaluation

The system can compare its performance on normal images and adversarially modified images to evaluate its robustness.

1. Flexible Model Selection

Different CNN architectures, Transformer architectures, datasets, and adversarial defence techniques can be selected based on the application.

1. Improved Reliability of Image Classification

By combining CNN-based feature extraction with Transformer-based relationship analysis, the invention aims to provide more reliable image classification under adversarial conditions.

1. Wide Range of Applications

The architecture can be adapted for security and surveillance, medical imaging, autonomous vehicles, facial recognition, industrial inspection, and agriculture.

1. **Has the invention been built or tested or implemented? If yes please provide the Efficiency/Effectiveness details of the invention**

No, the complete invention has not yet been fully implemented and tested.

The closest prior art is a hybrid CNN-Transformer architecture for image classification, which combines local feature extraction with global relationship analysis.

The proposed Hybrid CNN–Transformer Architecture has been designed and its working methodology has been defined. The proposed system includes image preprocessing, CNN-based feature extraction, Transformer feature processing, feature combination, classification, and adversarial testing.

A complete experimental implementation is planned using a suitable image dataset and a deep-learning framework such as Python with PyTorch or TensorFlow.

After implementation, the efficiency and efficacy of the proposed system can be evaluated using parameters such as:

- Accuracy
- Precision
- Recall
- F1-score
- Classification performance on normal images
- Classification performance on adversarial images
- Comparison with conventional CNN and Transformer models

Since complete experimental testing has not yet been performed, numerical performance results are not available at this stage.

1. Briefly state when and how you first conceived this idea?

The idea was conceived during the development of the assigned project on “Hybrid CNN–Transformer Architecture for Adversarial Image Classification.” The concept was developed by studying the capabilities and limitations of current image-classification techniques.

1. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have disclosed your invention. Were non-disclosure documents signed prior to disclosure in each case? Please provide the deadlines of which you may be aware for filing an application on this invention.

No. The proposed invention has not been sold, offered for sale, publicly used, or commercially published.

The invention has been developed as part of an academic project/assignment. No commercial activity has been carried out in relation to the invention.

The invention has not been formally disclosed to any external individual or organization for commercial purposes. Therefore, no non-disclosure agreement (NDA) has been signed in connection with such disclosure.

No specific deadline for filing a patent application is known at this stage. Any future patent filing will be conducted according to the applicable institutional and legal requirements.

1. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

The proposed invention is not merely based on the use of a CNN or a Transformer individually, as both techniques have been widely known and used in the field of image classification.

are already known for image classification. The proposed approach combines CNN-based local feature e
Transformer-based analysis of relationships between different parts of an image and applies this combine
adversarial image classification.

The invention integrates multiple stages, including image preprocessing, CNN feature extraction, feature
representation, Transformer processing, attention-based analysis, feature combination, adversarial trainin
classification, and robustness evaluation.

This combination is intended to address the limitations of using a single type of model. The CNN provides
local visual information, while the Transformer provides broader contextual information. The integrated ar
therefore designed to improve the reliability of image classification when images are intentionally modified

The specific combination of these elements for adversarial image classification provides a distinct approa
with conventional CNN-based classification, Transformer-based classification, or adversarial training use
independently.

1. **Additional comments by inventor (if you want to give more details out of scope of this IDF).**

--

1. **Drawings/Flowchart/Table**

